SYSTEMATIC REVIEW

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Efficacy and safety of wet wrap therapy for patients with atopic dermatitis: a systematic review and meta-analysis*

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Summary

Wet wrap therapy (WWT) consists of topical steroids administered under a layer of wet cotton bandages or garments. Several trials with WWT have reported promising results in atopic dermatitis (AD). However, no systematic review and meta-analysis on its efficacy and safety has been published. Our objective was to conduct a systematic review of the literature on WWT in AD to assess its efficacy and safety. We included randomized controlled trials among patients of all ages with a diagnosis of AD based on predefined criteria or made by a dermatologist. Electronic searches were performed from 1970 to 30 March 2016 in the Cochrane Central Register of Controlled Trials, MEDLINE, Embase and the World Health Organization International Clinical Trials Registry. Selection of studies and data extraction were performed independently by two researchers, and discrepancies were resolved by consensus. Six trials comparing WWT with topical steroids in children or adults with AD were included. Sample sizes ranged from 19 to 51 patients. Results on clinical severity and quality of life were reported incompletely and proved heterogeneous across studies. A nonsignificant tendency to increased risk of mild skin infections was observed in those treated with WWT (pooled relative risk 6·35, 95% confidence interval 0·83–48·55). The overall grade of quality of evidence for the efficacy and safety outcomes was low. In conclusion, the evidence that WWT is more effective than conventional treatment with topical steroids in AD is of low quality. Further clinical trials should establish the efficacy of WWT in AD.

What's already known about this topic?

- Several clinical trials, as well as some qualitative reviews, have reported promising results for wet wrap therapy (WWT) in the short-term management of severe flares of atopic dermatitis (AD).

What does this study add?

- This is the first systematic review on WWT for AD: it shows there is only low-quality evidence that this therapy is more effective than standard treatment with topical steroids.

Management of atopic dermatitis (AD) is based on the regular use of emollients, together with topical steroids or calcineurin inhibitors for acute flares. Phototherapy and immunosuppressive drugs are used in refractory cases.[1,2]

In most patients, AD flares can be controlled with topical therapy; however, some cases are unresponsive. Oral steroids are especially indicated for rapid control of severe or refractory disease.[2] Unfortunately, the risk–benefit balance of oral

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steroids is unclear, mainly due to their systemic adverse effects and the rebound phenomenon following treatment cessation.[2]

Wet wrap therapy (WWT) involves the application of a topical steroid, diluted or not, under two layers of cotton bandages or garments: a wet inner layer and a dry outer layer. Wraps are left for 8–24 h per day, and treatment duration should not exceed 14 days.[3,4] WWT is an interesting alternative for short-term control in patients with severe or refractory flares that avoids the use of systemic and potentially more toxic treatments.[3,4] Its efficacy has been claimed to derive from better penetration of the steroid, decrease in transepidermal water loss, or from the wraps acting as a barrier against scratching.[3] Several studies and qualitative reviews on WWT have reported promising results in the management of AD, although the number of patients included has been small.[5,6]

WWT has also been claimed to be safe, although its potential adverse effects include systemic absorption of the steroid, bacterial skin infections and discomfort associated with the procedure.[3,7] However, no systematic review and meta-analysis on the efficacy and safety of WWT has been published. Therefore, the aim of this study was to systematically review the literature to assess the efficacy and safety of WWT for patients with AD.

Materials and methods

Criteria for study selection

Randomized controlled clinical trials including patients of all ages with a diagnosis of AD made according to predefined criteria or by a dermatologist were included. WWT, defined as the application of a diluted or undiluted steroid under a layer of wet bandage or garment, should be compared against no treatment, placebo or other topical or systemic treatment for AD. The primary review outcome was clinical severity, and the secondary outcomes were quality of life (QoL), mild local adverse events (folliculitis and infected eczema) and severe adverse events (eczema herpeticum, irreversible skin changes such as atrophy or striae distensae and systemic steroid effects). For the timing of outcome assessment, we chose the available end point closest to 2 weeks. The reason for having chosen this timing is that WWT is mainly recommended for the short-term management of severe flares of AD.[1,3]

Search strategy

We searched the following electronic databases from 1970 to 30 March 2016: the Cochrane Central Register of Controlled Trials, MEDLINE through PubMed, Embase and the World Health Organization International Clinical Trials Registry Platform (http://apps.who.int/trialsearch/). No filters or language restrictions were used. Search strategies are reported in Table S1 (see Supporting Information).

We also checked the reference lists of included studies for further references to relevant trials. Finally, we searched the website www.opengrey.eu/ and established contact with researchers in the field to identify unpublished additional studies.

Data collection

After removing duplicate records titles and abstracts were screened to select the relevant reports. The full text of these studies was obtained and those meeting the elegibility criteria were included in the review. For each study we extracted the following information: study design, participants, interventions, outcome measures, results and risk of bias.

Assessment of risk of bias in each study was performed with the tool developed by the Cochrane Collaboration.[8] Each of the six domains in the tool was summarized as having low, unclear or high risk of bias. Quality of evidence on the effect of WWT on each clinical outcome was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system as high, moderate, low or very low, considering within-study risk of bias, heterogeneity, directness of evidence, precision of effect estimates and risk of publication bias.[8]

Selection of studies, data collection and grading of overall evidence were performed independently by two researchers (G.G.-L. and R.M.C.-R.), and discrepancies were resolved by consensus.

Statistical analysis

For continuous outcomes the effect of WWT was expressed as standardized mean difference (SMD) with its 95% confidence interval (CI). For dichotomous outcomes, the risk ratio (RR) and the corresponding 95% CI were used.

Heterogeneity of results across studies was tested with the Cochran Q test, and was quantified using the I^2 statistic.[9] A value of $I^2 > 50\%$ was considered as substantial heterogeneity. Characteristics of the studies were assessed in an attempt to understand the sources of heterogeneity. A pooled estimate of the WWT effect for each outcome was calculated when heterogeneity and clinical diversity across studies was deemed to be not too large. A random-effects model was chosen for $I^2 > 50\%$, and a fixed-effects model for $I^2 \leq 50\%$.

For missing data, we contacted the study authors for additional information. In case of nonresponse, we imputed the missing data when possible: for continuous outcomes we used the last observation carried forward (LOCF) method, and for dichotomous outcomes, we assumed that participants with incomplete data did not suffer the event.

Lastly, we performed sensitivity analyses, which consisted of excluding each study from the analysis, excluding studies at high risk of bias, exploring the effect of the type of meta-analysis used, assuming that missing data did suffer the outcome event and exploring the effect of the LOCF imputation. The small number of studies included in the review precluded the use of funnel plots to assess the risk of reporting bias.

Statistical analyses were performed with Review Manager 5.3 software.[10]

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Results

After excluding duplicates a total of 183 records were found. Of these, after assessing titles and abstracts, 26 records were selected for full-text assessment. Finally only six studies met all inclusion criteria and were selected for this review.[11–17] The reasons for exclusion of the remaining 16 studies are shown in Figure 1. The full text of one of these 16 studies could not be retrieved;[18] however, from reading the abstract it was considered unlikely to meet the inclusion criteria as it seems to be a nonrandomized study and no steroid cream seems to have been applied under the wraps. The characteristics of the six included studies are summarized in Table 1 (and Table S2; see Supporting Information).

Foelster-Holst *et al.*[13] randomized interventions to the left or right side of the body, so that each participant was its own control. The remaining studies were parallel trials. Sample sizes ranged from 19 to 51 patients.[12]

With regard to the diagnosis of AD, two studies used the Hanifin and Rajka criteria[14] or a modified version.[16] Beattie and Lewis-Jones[11] and Bryden *et al.*[12] focused on patients with mild or moderate AD whereas the rest included participants with moderate to severe disease. Only one trial included adults.[11] All participants in the other studies were children; in three of those studies all children were < 5 years old.[11,12,14]

All studies compared WWT against topical steroids with no wet wraps (conventional therapy). Two studies[11,12] compared a 2-week course of WWT and 1% hydrocortisone (HC) applied twice daily vs. 1% HC with no wet wraps twice daily. Foelster-Holst *et al.*[13] compared WWT and prednicarbate vs. prednicarbate alone, for a period of 48–72 h, although they did not specify the concentration of the steroid nor for how long the bandages were worn every day. Hindley *et al.*[14] compared WWT and 1% HC 24 h per day for 4 weeks vs. 1% HC alone twice daily, although after the first week the wet wraps could be worn overnight alone depending on the clinical response. Murphy[15] did not specify method or duration. Finally, Pei *et al.*[16] compared WWT with 1 : 10 diluted fluticasone propionate 0·005% or mometasone furoate 0·1% ointment on affected areas overnight for 2 weeks, vs. the same topical steroids alone.

All of the included studies addressed our primary outcome, clinical severity, which was measured by different objective scales (Table 2): Six Area, Six Sign Atopic Dermatitis severity score (SASSAD),[11,12,15] local SCORing Atopic Dermatitis (local SCORAD),[13] SCORAD[14] and an objective ad hoc scale.[16] Several studies addressed our secondary outcomes, namely

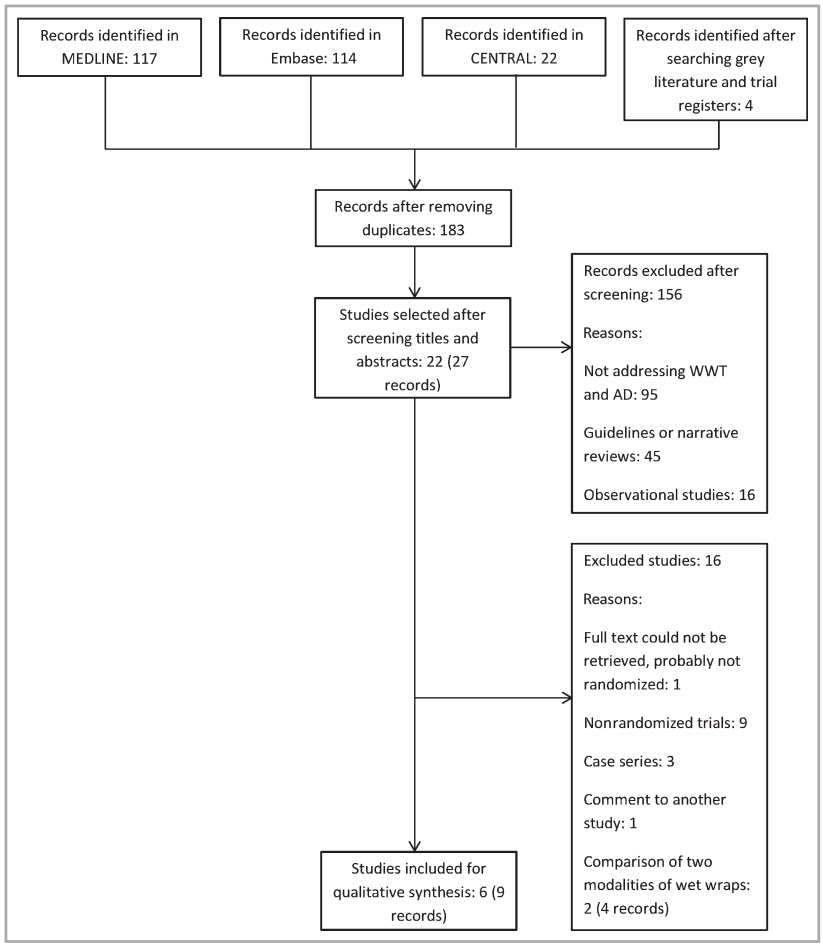

Fig 1. Flow diagram of studies across the review. AD, atopic dermatitis; CENTRAL, Cochrane Central Register of Controlled Trials; WWT, wet wrap therapy.

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Table 1 Characteristics of included studies

Study	Design	n	Participants	Intervention	Duration of intervention	Outcomes
Beattie 2004[11]	RPT	19	Age < 5 years; BSA > 30%	Intervention: WWT with HC 1% twice daily then overnight the second week. Control: HC 1% twice daily	2 weeks	SASSAD, IDQOL, DFI, observed adverse events
Bryden 2009[12]	RPT	51	Age < 5 years; mild to moderate AD	Intervention: WWT with HC 1% twice daily then once daily the second week. Control: HC 1% twice daily	2 weeks	As in Beattie 2004[11]
Foelster-Holst 2006[13]	Randomized trial with left –right comparison	24	Age 6–63 years; local SCORAD > 10, similar on both sides	Intervention: WWT with prednicarbate Control: prednicarbate	48–72 h	Local SCORAD, observed adverse effects
Hindley 2006[14]	RPT	50	Age 3 months–5 years; SCORAD > 15	Intervention: WWT with HC 1% 24 h. It could be reduced to 12 h per day after the first week. Control: HC 1% twice daily	4 weeks	SCORAD, observed adverse effects
Murphy 2003[15]	RPT	36	Age: 5 months–9 years; moderate to severe AD	Intervention: WWT with mild steroid. Control: mild steroid	Not reported	SASSAD, GOS score[a]
Pei 2001[16]	RPT	28[b]	Age: 1–15 years; severe AD refractory to potent steroids	Intervention: WWT with 1 : 10 dilution of potent steroid overnight. Control: 1 : 10 dilution of potent steroid once daily	2 weeks	Clinical severity score, disease extent, QoL questionnaire

RPT, randomized parallel trial; BSA, body surface area; WWT, wet wrap therapy; HC, hydrocortisone; SASSAD, Six Area, Six Sign Atopic Dermatitis severity score; IDQOL, Infant's Dermatitis Quality of Life Index; DFI, Dermatitis Family Impact; SCORAD, SCORing Atopic Dermatitis. [a]GOS score not defined in Murphy *et al.*[15] [b]The study consisted of four arms as participants were first randomized to mometasone furoate or fluticasone propionate, and then to wet wraps. Only the comparison between those that used wet wraps and those who did not is taken into account for this review.

QoL[11,12,16] and mild local adverse effects;[11,13,14] however, no study reported severe adverse events.

Risk of bias in the reviewed studies

The assessment of risk of bias is presented in detail in Figure 2 (and Table S3; see Supporting Information). Only three studies[11,12,14] described an adequate randomization method. All studies except two[11,13] reported blinding of the assessor. Another important source of potential bias was the incomplete reporting of outcome data in the WWT group by Beattie and Lewis-Jones[11] and Hindley *et al.*[14] A LOCF imputation was performed for the missing severity data in the former, while a per-protocol analysis was done in the latter.

Efficacy and safety of wet wrap therapy

Differences in clinical severity could be calculated for three studies and expressed as SMD (Fig. 3). Due to the high heterogeneity of results across studies a pooled estimate was not calculated. The LOCF imputation was only applied in Beattie and Lewis-Jones; when the sensitivity analysis assuming an available-case scenario was done the results proved similar, which indicates that the imputation method does not materially affect the results (Table S4; see Supporting Information). As for those studies not included in Figure 3, one found a significant improvement with the WWT group but did not report its magnitude,[15] another observed a nonsignificant difference in the median reduction of severity of −2·5 (95% CI −6 to 7) that favoured WWT,[12] and another study found a significant clinical improvement under WWT but its magnitude was not reported.[16]

With regard to QoL Beattie and Lewis-Jones found nonsignificant differences in the Infant's Dermatitis Quality of Life Index (IDQOL)[19] (2·1, 95% CI −1·62 to 5·82) and Dermatitis Family Impact (DFI)[20] (1·3, 95% CI −3·50 to 6·10).[11] Pei *et al.* reported that change from baseline in the QoL score was statistically significant in the WWT group but not in the group with topical steroids; however, they did not report a direct comparison between both treatments.[16] Bryden *et al.* measured but did not report the values of IDQOL and DFI.[12]

Three studies reported local adverse events:[11,13,14] Hindley *et al.* observed five of 28 cases of infected eczema in the WWT group vs. zero of 22 in the topical steroids group[14] and

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Table 2 Outcome measures used in the included studies

Scale	Study	Definition	Range	Validated
SASSAD	Beattie 2004,[11] Bryden 2009,[12] Murphy 2003[15]	Six signs are graded from 0 to 3 (erythema, oozing, excoriation, dryness, cracking and lichenification) at six sites: arms, hands, legs, feet, trunk, and head and neck. A higher number implies higher severity	0–108	Yes[26]
SCORAD	Hindley 2006[14]	Six signs from a representative site are graded from 0 to 3 (erythema, oedema, oozing/crusting, excoriations, lichenification and dryness): sum = B. Calculated percentage of body surface affected = A. Itch and lack of sleep are assessed from 0 to 10: sum = C. SCORAD = A/5 + 7B/2 + C. A higher number implies higher severity	0–103	Yes[27]
Local SCORAD	Foelster-Holst 2006[13]	The six SCORAD signs from a representative site are graded from 0 to 3 and added. A higher number implies higher severity	0–18	No
Severity scale	Pei 2001[16]	The six SCORAD signs are graded from 0 to 3, in eight body sites: head and neck, anterior trunk, back, genitals and the four extremities. A higher number implies higher severity	0–144	No
IDQOL	Beattie 2004,[11] Bryden 2009[12]	Ten questions on itch, lack of sleep, mood and daily activities are graded from 0 to 3. Valid for children < 4 years old. A higher number implies a lower QoL	0–30	Yes[18]
DFI	Beattie 2004,[11] Bryden 2009[12]	Ten questions on the impact the disease has on the patient's parents are graded from 0 to 3. A higher number implies a higher impact	0–30	Yes[19]
QoL questionnaire	Pei 2001[16]	Questions on itch, pain and the impact of disease on sleep and daily activities are graded from 0 to 3. A higher number implies a higher impact	Not specified	No

SASSAD, Six Area, Six Sign Atopic Dermatitis severity score; SCORAD, SCORing Atopic Dermatitis; IDQOL, Infant's Dermatitis Quality of Life Index; DFI, Dermatitis Family Impact.

Beattie and Lewis-Jones found two of 10 cases of folliculitis in the WWT group vs. zero of nine in the topical steroids group.[11] A meta-analysis of these results using a random-effects model showed a nonsignificant tendency to higher risk of local adverse events in the WWT group (RR 6·35, 95% CI 0·83–48·55; Fig. 3). However, this finding achieved statistical significance when the analysis assumed that patients with missing data did experience local adverse events (RR 16·66, 95% CI 1·63–83·96) (Table S4; see Supporting Information). When we excluded the studies at high risk of bias, no studies remained in the analysis. Thus, no overall estimation could be made in this case.

Discussion

Overall, three studies detected a significant improvement in clinical severity favouring WWT vs. conventional treatment, whereas the other three studies obtained nonsignificant results. Interestingly, one included study[14] reported a slight, nonsignificant worsening in disease severity in the WWT group at week 4, which could point to a decrease in efficacy after the first few weeks. As for QoL, Beattie and Lewis-Jones[11] found a small, nonsignificant difference unfavourable to WWT, whereas the other studies either did not address this outcome or reported it insufficiently. Finally, the difference in mild local skin infections between the two groups was not statistically significant.

There were also some limitations in the type of information yielded by the reviewed studies. Firstly, duration of treatment and steroid potency were highly variable in the included studies, though most used mild-potency steroids. Secondly, the results of this review apply only to the relative efficacy of WWT vs. topical steroids as there were no randomized trials comparing WWT with other treatment modalities for AD. Thirdly, most of the studies included children, and only one included adults:[13] interestingly, this was one of the trials where the results were most favourable to WWT. Fourthly, one of the trials was a left–right study,[13] which makes it impossible to evaluate outcomes such as QoL and systemic side-effects. Fifthly, although the reviewed studies did not observe systemic or severe cutaneous events, such as growth retardation or hypothalamic–pituitary–adrenal axis suppression, such events were not actively investigated. Sixthly, the small sample sizes of the studies allowed for detecting only frequent adverse outcomes. Finally, the short follow-up time, which was not longer than 4 weeks in any of the trials, further limited detection of these events. Thus, the evidence from these studies cannot be used to draw definitive conclusions in this regard.

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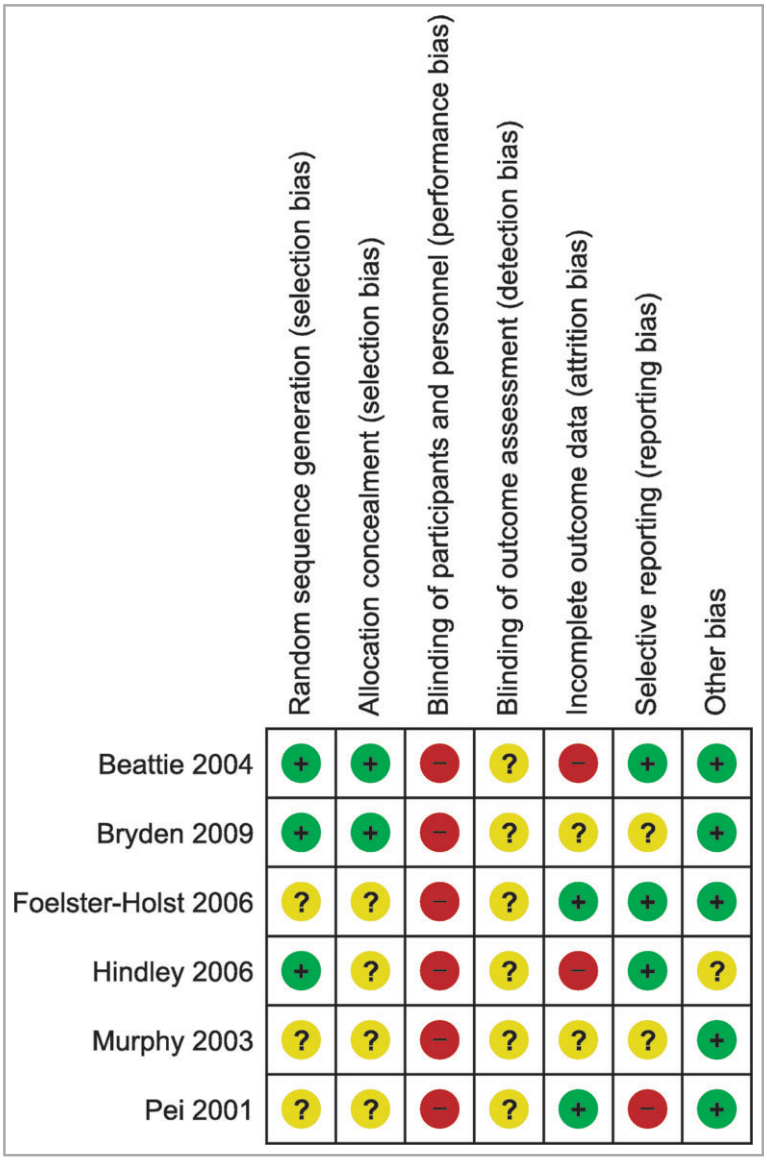

Fig 2. Risk of bias of the included studies. The domain 'Blinding of outcome assessment' was considered to be at high risk of bias for the quality of life outcome as patients were their own assessors.

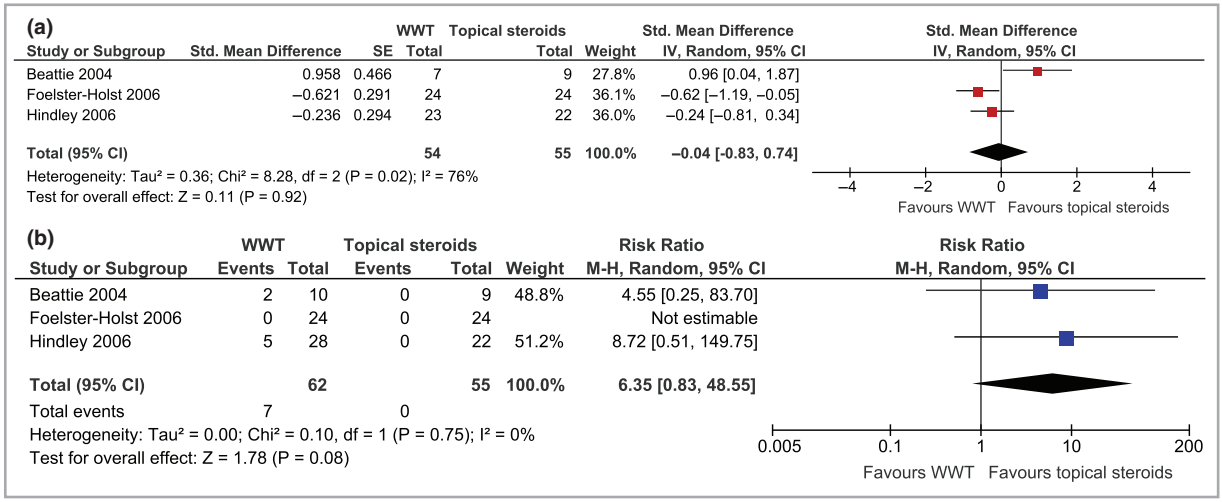

Fig 3. Forest plot of trials comparing wet wrap therapy (WWT) vs. topical steroids. (a) Outcome: clinical severity. (b) Outcome: local adverse events. CI, confidence interval. M-H refers to the Mantel-Haenszel test, which was the statistical method used for the meta-analysis of mild adverse events.

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Table 3 Quality of the evidence and summary of findings of wet wrap therapy (WWT) vs. topical steroids (TS) on several outcomes among patients with atopic dermatitis

Quality assessment							Number of patients		Effect			
Number of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	WWT	TS	Relative (95% CI)	Absolute (95% CI)	Quality[a]	Importance
Clinical severity												
6	Randomized trials	Serious[b]	Serious[c]	Not serious	Not serious	None	104[d]	92[d]	Not pooled[f]	Not pooled[f]	⊕⊕◯◯ Low	Critical
Quality of life												
3	Randomized trials	Serious[b]	Not serious	Not serious	Serious[e]	None	52	46	Not pooled[f]	Not pooled[f]	⊕⊕◯◯ Low	Critical
Mild local adverse events												
3	Randomized trials	Serious[b]	Not serious	Not serious	Serious[g]	None	7 of 62 (11%)	0 of 55 (0%)	RR 6·35 (0·83–48·55)	11 more per 100 (from 8 fewer to 30 more)[h]	⊕⊕◯◯ Low	Important

CI, confidence interval; RR, risk ratio. [a]Grading of Recommendations Assessment, Development and Evaluation system. [b]Because all the reviewed studies had a 'high risk' or 'unclear risk' of bias judgement on at least one of the risk-of-bias items (Fig. 2 and Table S3; see Supporting Information). [c]Due to high heterogeneity, which precluded meta-analysis. [d]Participants in Murphy *et al.*[15] have not been counted, as the number of patients assigned to each group is not reported. [e]Results on quality of life could only be obtained from a small number of patients: only Beattie and Lewis-Jones[11] provide a direct comparison and its results have wide CIs (see text). [f]A pooled estimate was not obtained for these outcomes (see text and Fig. 3). [g]Results on mild adverse events could only be obtained from a small number of participants, CIs are wide and include both higher and lower incidence of events in the WWT group (Fig. 3). [h]Absolute risk was obtained by calculating the pooled risk difference in the meta-analysis of mild local adverse events.

According to the GRADE system, overall quality of the evidence was low for all the outcomes considered for this review (Table 3). The results of Beattie and Lewis-Jones[11] and Hindley *et al.*[14] might have a high risk of attrition bias; the remaining studies have unclear bias in several items of the Cochrane tool. In addition, the small sample sizes in the trials led to imprecise results with wide CIs, which further limits the quality of the evidence.

We found high heterogeneity among studies in the results for clinical severity, which precluded calculating a pooled estimate of the effect of WWT. Heterogeneity could be partly due to the diversity of study participants, treatment regimens and scales used.

With respect to potential biases in the review process, given that bibliographic searches were performed in several databases and that grey literature was also searched, it is unlikely that we missed relevant studies.[5,6,21,22] However, the small number of included studies made the funnel plots meaningless, and no conclusions on publication bias could be drawn. Finally, the imputation methods used can be another source of bias. Changing the imputation assumptions in the meta-analysis of local adverse events had a relevant influence on the pooled effect, but this issue has been dealt with by including both scenarios in this article.

Comparing this with other studies, our results are consistent with those of other reviews.[5,6,21] However, previous reviews did not attempt a quantitative synthesis of data and also included nonrandomized studies and case series. Perhaps because of this, one review[6] concluded that WWT is effective for severe cases of AD but acknowledged that the level of evidence was weak.

Some studies describe infectious events such as herpes simplex infection or upper respiratory tract infection, which were not reported in the studies included in this review.[23] In contrast, another study[24] found no cases of local infection in 217 children treated with wet wraps, although the mean duration of treatment was 3·6 days. It is likely that a short duration implies a lower infection risk, which is consistent with the findings by Foelster-Holst *et al.*[13] Other studies have found a suppression of the hypothalamic–pituitary–adrenal axis in some patients[23,25] but they did not further investigate its clinical consequences.

Looking at the implications for practice, our review found the evidence that WWT has a greater efficacy than topical steroids, for the treatment of AD is of low quality. The effect of treatment on QoL was also inconclusive.

WWT may lead to a higher risk of mild local infections (folliculitis and infected eczema) than topical steroids, but again, this could not be proven. No cases of severe cutaneous or systemic events were found.

With respect to implications for research, we believe that methodologically rigorous and adequately reported randomized trials with a higher number of participants are needed to establish the efficacy of WWT vs. topical steroid treatment in AD. Validated scales for clinical severity and QoL should be used, and adverse effects should be actively investigated and

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reported. Moreover, studies with a longer follow-up are needed for registering systemic effects of steroids, as well as establishing the duration of treatment response.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Table S1. Bibliographic electronic search strategy.
Table S2. Characteristics of the included studies.
Table S3. Risk of bias of the reviewed studies.
Table S4. Sensitivity analysis.
PowerPoint S1. Author video.

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